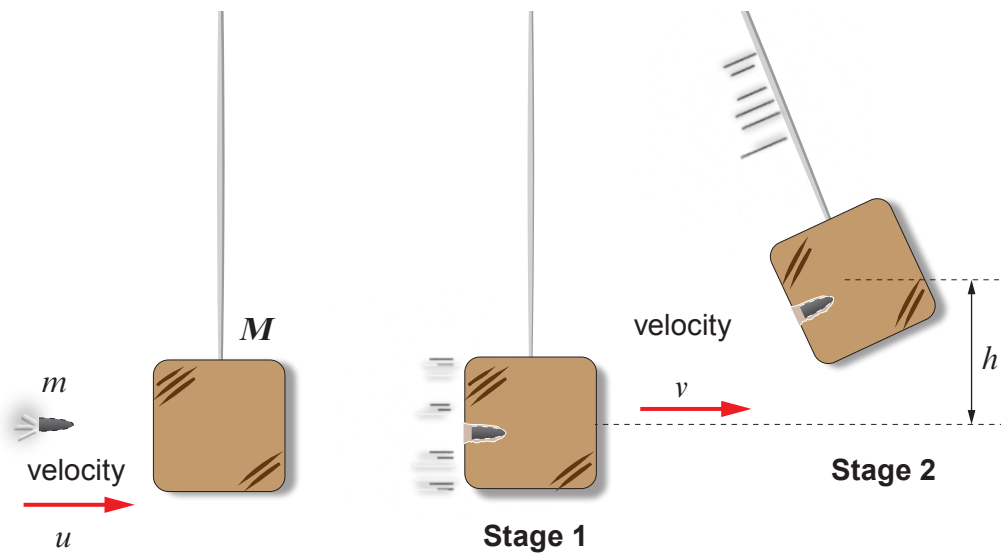


7. The velocity of a bullet can be determined by firing it into a suspended wooden block. As the bullet enters the block it will cause it to swing through a height, h . This is shown in the diagram below.



- (a) (i) State the law of conservation of momentum. [2]

.....

.....

.....

.....

- (ii) Explain whether this law applies to the block as it swings up. [1]

.....

.....

- (b) Discuss whether kinetic energy is conserved in **both** stages of the process. [2]

.....

.....

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- (c) The mass of a bullet is 16 g. It is fired at a velocity, u into a stationary wooden block of mass 2.5 kg. The combined bullet and block start to move with a velocity, v . The block swings through a height, h , equal to 11.9 cm before it comes to rest again.

Use the information above and equations from page 2 to answer the following questions.
($g = 10 \text{ m/s}^2$)

- (i) Calculate the velocity, v . [4]

Velocity, $v = \dots\dots\dots \text{ m/s}$

- (ii) Use your answer in part (i) to calculate the velocity, u , of the bullet. [4]

Velocity, $u = \dots\dots\dots \text{ m/s}$

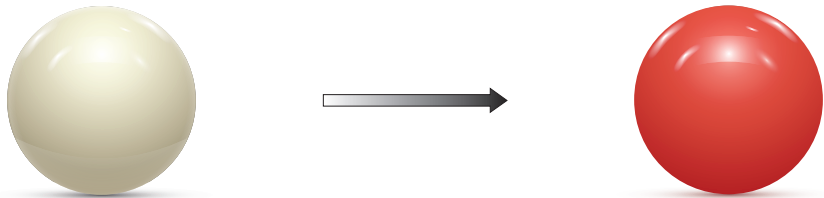
13

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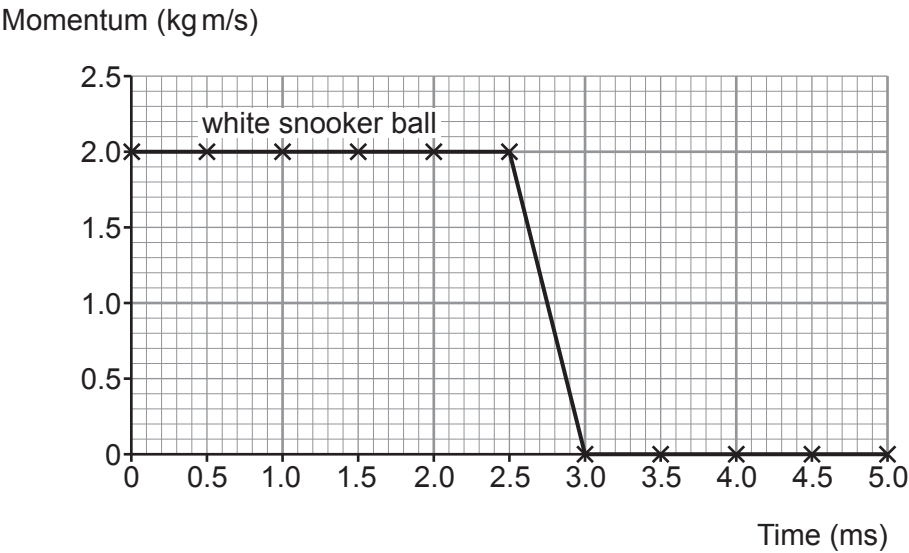


Answer **all** questions.

1. A white snooker ball, rolling to the right, collided with a **stationary** red snooker ball at a time of 2.5 ms.
Each snooker ball has a mass 0.16 kg.



The graph below shows the momentum of the **white snooker ball** before and after the collision.



- (a) Use information from the graph to answer the following questions.

- (i) State the initial momentum of the white snooker ball. [1]

initial momentum = kg m/s



- (ii) Use the equation:

$$\text{initial velocity} = \frac{\text{initial momentum}}{\text{mass}}$$

to calculate the initial velocity of the white snooker ball.

[2]

initial velocity = m/s

- (iii) The collision takes a time of 0.5 ms. State this time in seconds.

[1]

time = s

- (iv) Use the equation:

$$\text{resultant force} = \frac{\text{change in momentum}}{\text{time}}$$

to calculate the resultant force on the white snooker ball during the collision.

[2]

resultant force = N

- (b) (i) Underline the correct word or phrase from each of the brackets below which correctly completes the sentence.

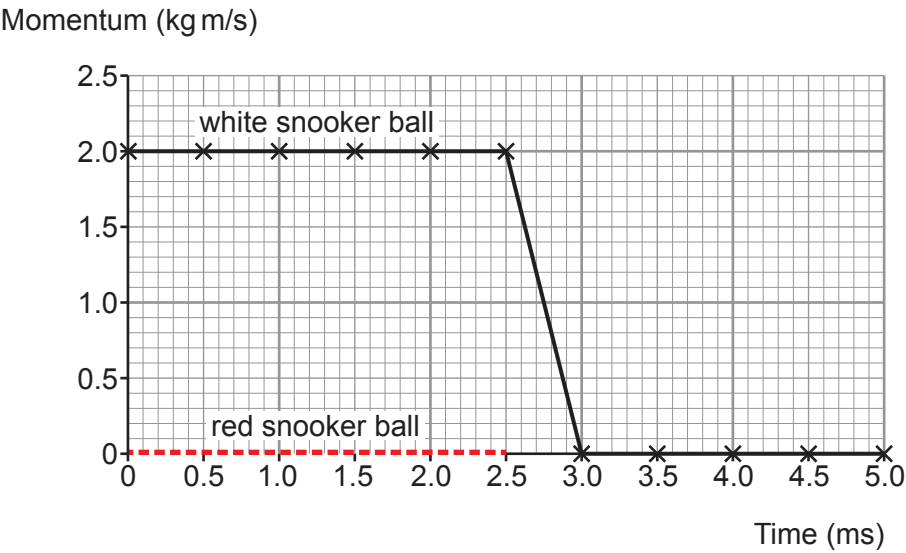
[2]

The law of conservation of momentum states that the total momentum before a collision is (**less than** / **equal to** / **greater than**) the total momentum after a collision provided (**no** / **small** / **large**) external forces act.



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- (ii) The momentum of the red snooker ball, between 0.0 and 2.5 ms, has been added to the original graph. It is shown as a red dotted line.
Complete the graph below to show the momentum of the red snooker ball from 2.5 ms to 5.0 ms. [2]



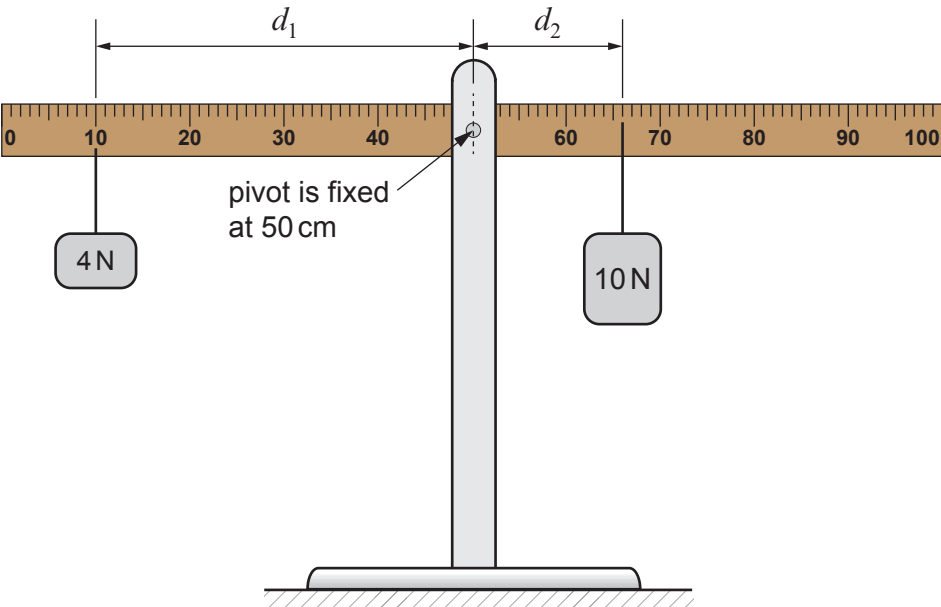
10

3420UD01
05



3. Mary investigated moments using a **100 cm** ruler and two different weights.

The apparatus used is shown below.



Mary checked the ruler was horizontal. She then attached weights to the ruler. The 4 N weight, W_1 , was attached to the ruler a distance, d_1 , from the pivot. The 10 N weight, W_2 , was then attached to the ruler to make it horizontal again. The distance, d_2 , of the 10 N weight from the pivot was noted.

Mary's results are shown in the table.

Weight, W_1 (N)	Distance, d_1 (cm)	Weight, W_2 (N)	Distance, d_2 (cm)
4	40	10	16
4	35	10	14
4	20	10	8
4	15	10	6
4	5	10	2

(a) Mary states the resolution of the ruler she used in this experiment as 1 cm. Use information from the diagram to explain whether Mary is correct.

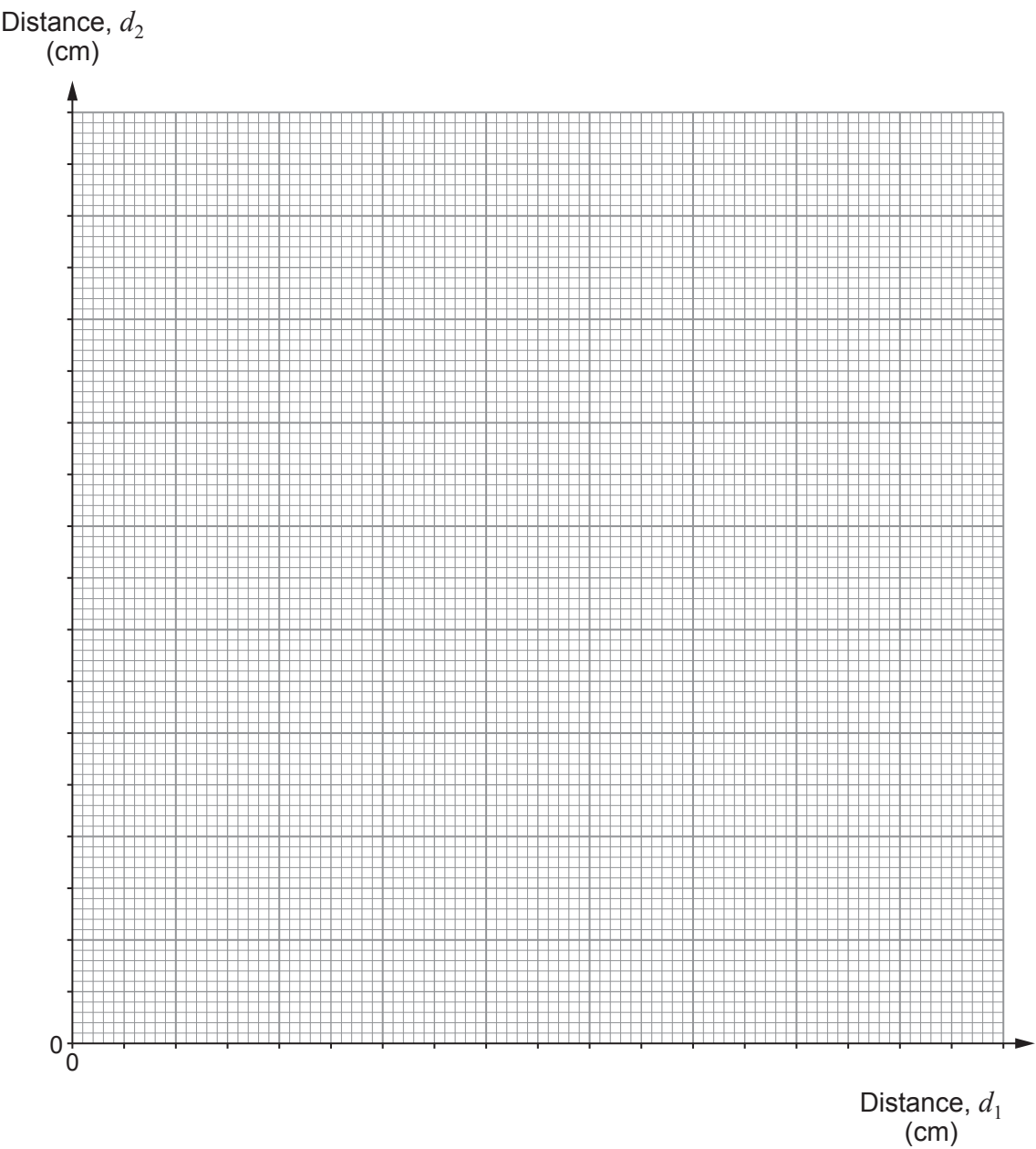
[1]



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(b) (i) Plot the data on the grid below and draw a suitable **straight** line.

[3]



3420UD01
11



Examiner
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(ii) Mary suggests that the value of the gradient of the graph is the same as $\frac{W_1}{W_2}$.

Use data from the graph and the table to explain whether Mary is correct. [3]

.....

.....

(c) (i) Mary now places the 10 N weight at a distance, d_2 , of 32 cm from the pivot.
Use an equation from page 2 to calculate its clockwise moment about the pivot. [2]

Clockwise moment =Ncm

(ii) Explain, using moments, why the ruler cannot now be balanced using the 4 N weight. [2]

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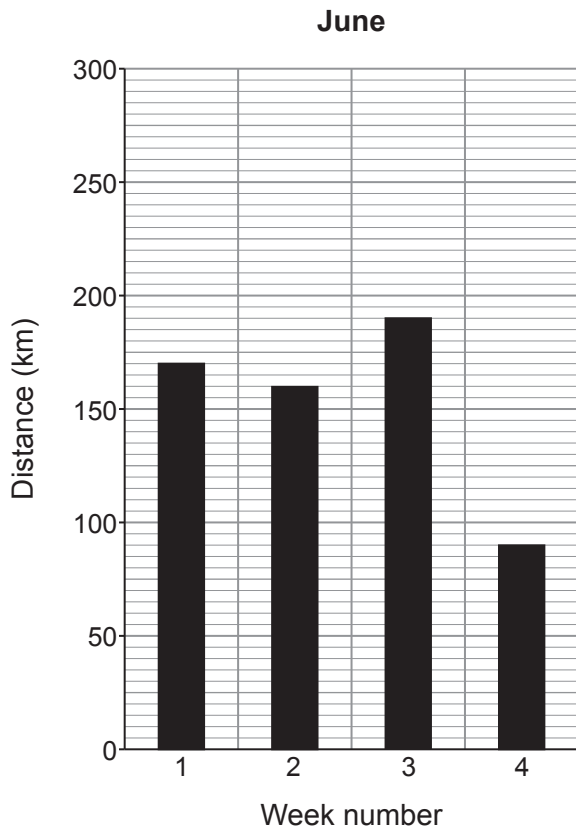
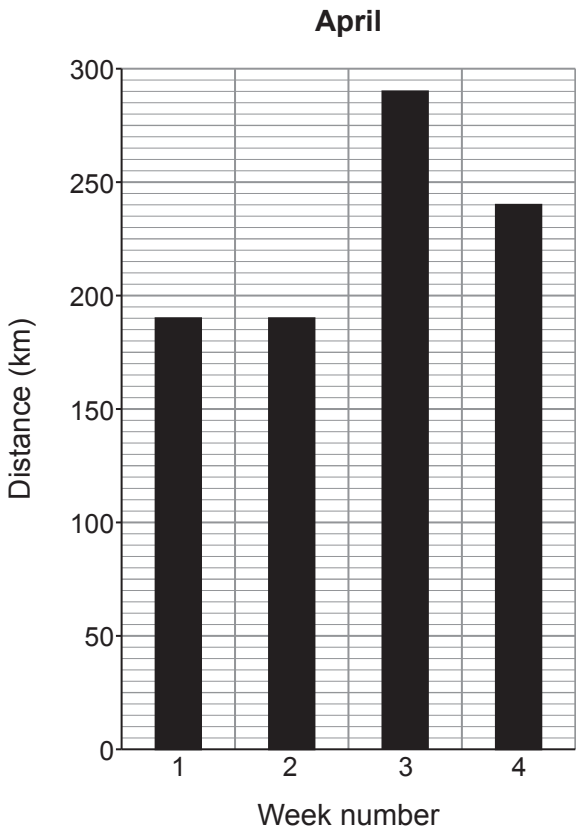
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11



4. A fitness app is used by a cyclist to monitor his cycling performance.

- (a) The distance cycled each week is displayed as a chart.
The data for two months, April and June, are shown below.



- (i) The total distance travelled by the cyclist in June was 610 km.
Show that the total distance travelled by the cyclist in April was 910 km.

[1]



- (ii) The cyclist cycled for a total time of 31.5 hours during April and 19.25 hours during June.
A student suggests that the mean speed of the cyclist is greater in June than in April.
Use an equation from page 2 to determine whether the student is correct. [3]
Space for calculations.

.....

.....

- (b) (i) A track cyclist has an initial velocity of 8.0 m/s.
She uniformly accelerates towards the finish line, travelling a distance of 42 m, in a time of 3.0 s. [4]

Use the equation:

$$x = ut + \frac{1}{2}at^2$$

to calculate the cyclist’s acceleration and state its unit.

acceleration =

unit =

- (ii) The resultant force that causes the acceleration towards the finish line is 284 N.
I. Use an equation from page 2 to calculate the mass of the bike and the cyclist. [2]

mass of bike and cyclist = kg



II. The mass of the cyclist is 64.5 kg. Calculate the mass of the bike. [1]

mass of bike = kg

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11



8. This question is about momentum.

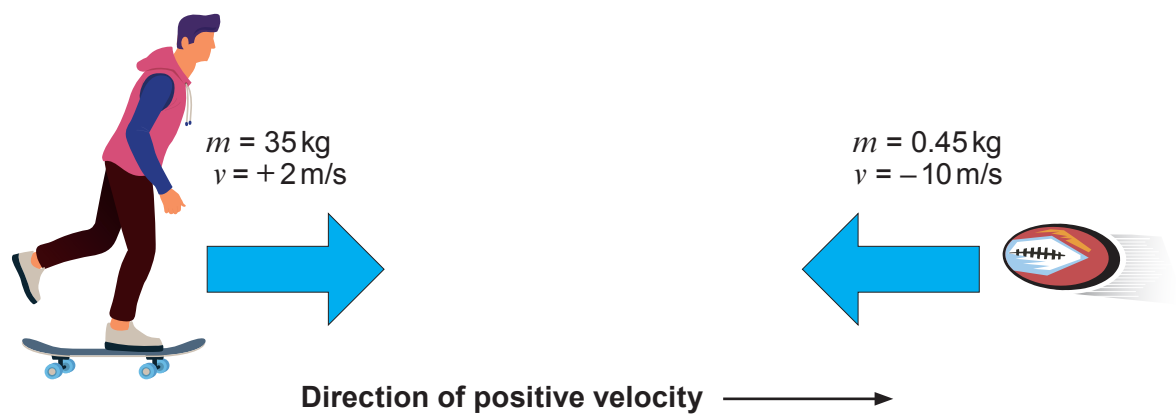
(a) State the principle of conservation of momentum. [2]

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.....

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(b) A skateboarder of mass 35 kg moving with a velocity of 2 m/s catches and holds on to a 0.45 kg rugby ball thrown directly towards him at 10 m/s.



Use the equation:

momentum = mass × velocity

to calculate the **total** momentum of the skateboarder and the ball before the collision and give the unit. [4]

Total momentum =

Unit =



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- (c) Use an equation from page 2 to calculate the velocity of the skateboarder after he catches the rugby ball. [2]

Velocity = m/s

- (d) Tomos suggests that **kinetic energy** is conserved in this collision. Explain, by using calculations, whether or not you agree. Use an equation from page 2. [4]

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12

